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Lectures

Pruning and sparsity

Which weight can go, and how the others repair the loss.

The question	Remove a weight while changing the loss as little as possible. The local model is a quadratic in the weights.
OBD	Treat the Hessian as diagonal. Saliency of a weight is $\frac{1}{2} H_{ii} w_i^2$. Nothing else is updated.
OBS	Use the full Hessian. Saliency is $w_i^2 / (2 (H^{-1})_{ii})$, and the other weights shift to cancel the damage.
The cost of H	A diagonal by backprop, Hessian–vector products, and a Sherman–Morrison inverse in the MSE case.
Retrain	If the best saliency is still large, the quadratic step is not enough and the network has to be trained again.

Pruning in practice

A zero is cheap only when the kernel can skip the work.

Structure	Dropped channels or layers stay dense. N:M sparsity, such as 2:4, is the pattern Ampere can accelerate.
Magnitude	Han’s recipe prunes, retrains, then quantizes and entropy-codes. Iterative magnitude pruning is the lottery-ticket loop.
SparseGPT	One shot, layer by layer. Drop a column and push the error into the columns still free, using a calibration Hessian.
Wanda	Score a weight by its magnitude times the activation norm on that input. No second-order update.
Bonsai	Forward passes only. Sample feasible sub-models, fit a score per module, and keep modules inside the size budget.

Sampling

Estimate an expectation, draw from a distribution, or keep a sample of a stream.

Why not uniform	A uniform average is unbiased and still useless when one term dominates. That is the usual shape of attention.
Importance	Draw from a proposal g and reweight by f/g. Unbiased for any g. Variance is smallest when g tracks |w| f.
Rejection	When the goal is a sample from f, not an average: draw from a g that covers f, and accept with probability f/(Mg).
Tail bounds	Markov, Chebyshev, then Chernoff from the moment generating function. These turn a sample size into an (ε, δ) guarantee.
Reservoirs	A stream is seen once. Uniform reservoir: a new item enters with probability k/(t+1). Weighted keys match sampling by weight.

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Quantization

Fewer bits, without letting one outlier set the scale for everyone.

The map	A scale, and a zero-point when the range is not centered. The codebook can be an integer grid or a floating-point format.
Formats	FP16, BF16, FP8, and block formats such as MXFP8 and NVFP4. The format pays off only when the kernel actually uses it.
Granularity	One scale per tensor, per channel, or per group. A finer scale survives outliers and costs more metadata.
Outliers	Clipping biases the rest of the vector. A random Hadamard rotation spreads the energy, in O(n log n), and is cheap to store.
Rounding	Round-to-nearest biases an inner product. Stochastic rounding is unbiased, with higher variance.

Quantizing a model

Where the error is allowed to go, and how training sees the quantizer.

PTQ and QAT	Post-training uses a calibration set. QAT inserts fake quantization and trains through it with a straight-through estimator.
GPTQ	The same second-order step as OBS. Quantize one column and compensate the columns that are still free.
AWQ	Activation size marks the channels that matter. Rescale those channels so they keep their precision.
SmoothQuant	Move activation outliers into the weights, so both sides of the product can use a low-bit format.
Codebooks	When one scale is not enough, product quantization gives each subspace its own codebook.

Projections

A short random map that keeps the geometry we actually use.

JL lemma	n points embed into $r = O(\log n / \varepsilon^2)$ dimensions with every distance preserved up to $1 \pm \varepsilon$. r does not depend on the original dimension.
The map	Gaussian entries of size $1/\sqrt{r}$. One pair concentrates, and a union bound covers every pair.
Regression	Preserving one residual is not enough. The sketch has to preserve the whole column space of A.
Subspace embedding	$(1-\varepsilon)\|\|Ax\|\| \le \|\|SAx\|\| \le (1+\varepsilon)\|\|Ax\|\|$ for every x. A net of the sphere reduces this to a finite JL problem.
CountSketch Matrix	One random ± 1 per column. As sparse as a subspace embedding can be, with r on the order of d^2/ε^2.

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People who made Class worthwhile for me.

Strong seminar talks

One seminar is still left. Advay Bapat's seminar was missed.

1

CLOSE TIE

Rishi Raj Vishwakarma (SOAR)

Apoorv Goyal (TurboQuant)

2

Tejas Chaudhary (SCANN)

3

Deeptanshu Malu (Sparse Prefill)

Who joined in during class?

From memory, so some names worth mentioning may be missing.

1

Ishan

2

Vedant

MANY OTHERS ENGAGED VERY WELL

Among them **Amit, Devansh, and Juhi**. The list is not complete.

Scribe support

SCRIBE

Amit Tiwari

(last few classes is entirely Amit's scribes)

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Seminars

Two topics. Decode-time sparse attention, and compressing the KV cache.

Sparse attention (decode)

The bottleneck

One new token against the whole cache. The step is a memory-bound GEMV, and exact k-sparse selection is NP-hard.

PQCache

Product-quantize the keys, score them with a lookup table, and fetch only the top-k full KV from memory.

HashAttention

A semantic hash chooses the tokens worth reading, instead of scanning every key.

MagicPIG

Write attention as an expectation and estimate it with LSH samples.

vAttention

Sinks, a local window, and top-k, plus a random sample sized for an (ε, δ) error bound.

KV-cache quantization

TurboQuant

Rotate, then quantize. An MSE codebook biases inner products; one bit on the residual corrects that.

KVTC

Undo RoPE, share a PCA across layers, and assign bits by dynamic programming. Sinks and recent tokens stay exact.

KIVI

Keys are quantized per channel, values per token. A short residual window stays in full precision.

KVQuant and CQ

Per-channel scales handle outliers. Coupled codes, one code for several channels, are what still work near 1 bit.

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Seminars

Two topics. Sparse attention while the prompt is read, and guessing the next tokens.

Sparse attention (prefill)

MInference

Choose a pattern per head: an A-shape, a vertical slash, or blocks. The dense matrix is never formed.

SpargeAttention

Skip blocks whose queries and keys cannot score high. The follow-up learns that pattern instead of only detecting it.

Sparse VideoGen2

Permute video tokens by semantics so attention lines up into blocks the kernel can skip.

DeepSeek

V3.2 and V4 build long-context attention into the model, rather than choosing a mask at inference.

Speculative decoding

The serial step

Scheduling can hide CPU work. It cannot make the next token exist before this forward pass finishes.

Leviathan et al.

A small draft proposes several tokens. One pass of the large model checks the whole draft.

Speculative sampling

Chen et al. accept or resample so the tokens that remain are distributed exactly as the target model.

In a server

vLLM still runs one batched forward pass. The draft changes how many tokens that pass is allowed to commit.

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Seminars

Two topics. Sharing parameters by hashing, and attention that is linear in length.

Projection learning

HashedNets

Chen et al. hash connections into buckets that share one weight. There is no index table to store.

ROBE

One circular array replaces the embedding tables. A hash reads a contiguous block.

ROAST

Hash a tile shaped like the matmul, so the load is coalesced instead of scattered.

STABLE-RPS

One shared array mixes the gradients. A per-slot scaler is what keeps training stable.

SS1

Store a sketch of each neuron. The product becomes a sketch of the input times a short vector.

Linear attention

Random features

Rahimi and Recht. A kernel machine becomes linear in the number of examples.

Transformers as RNNs

Katharopoulos et al. A factorable kernel turns attention into a recurrent state of fixed size.

Performers

Choromanski et al. Positive random features stand in for softmax, and the cost is linear in length.

Reformer

Kitaev et al. Locality-sensitive hashing chooses which pairs interact. Reversible layers hold the memory down.

Linformer

Wang et al. Project keys and values along the sequence. Attention then runs on a short, low-rank axis.

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Seminars

Maximum inner product search. Routing and caching is next Friday.

Approximate nearest neighbours

PQ

Split each vector into subspaces and store a code per subspace. A query is scored by table lookup.

ScaNN

The codebook charges more for error along the inner product. Orthogonal error is cheaper to leave.

SOAR

A point near a partition boundary gets a second copy, using a residual orthogonal to the first assignment.

HNSW

A stack of proximity graphs. The walk starts on the sparse top layer and drops to the dense bottom.

DiskANN

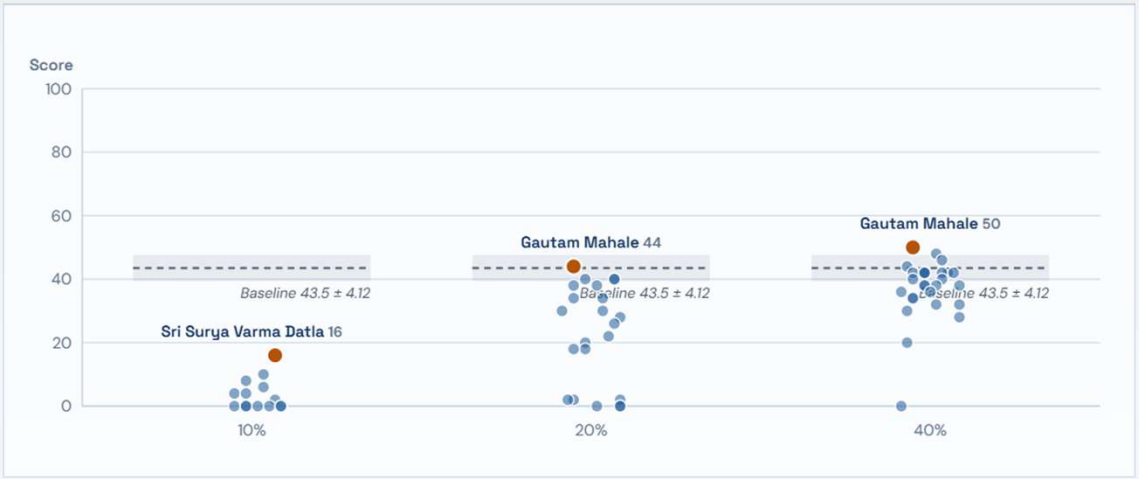
Compressed vectors in memory guide the walk. Full-precision vectors stay on disk and are read for a short list.

Routing / caching

Next Friday

The papers for this seminar are not chosen yet.

Project (leaderboard on website)



10% compression

RANK	ALIAS	SCORE
—	*Baseline*	*43.5 ± 4.12*
1	Sri Surya Varma Datla	16
2	Rishit Aggarwal	10
3	Siddharth N	8
4	Prakhar Paliwal	6
5	Amit Tiwari	4
5	Dhruv Savot	4

20% compression

RANK	ALIAS	SCORE
1	Gautam Mahale	44
—	*Baseline*	*43.5 ± 4.12*
2	Dhruv Savot	40
2	Kislay Aditya Oj	40
2	Sri Surya Varma Datla	40
5	Advay Bapat	38
5	Apoorv Goyal	38

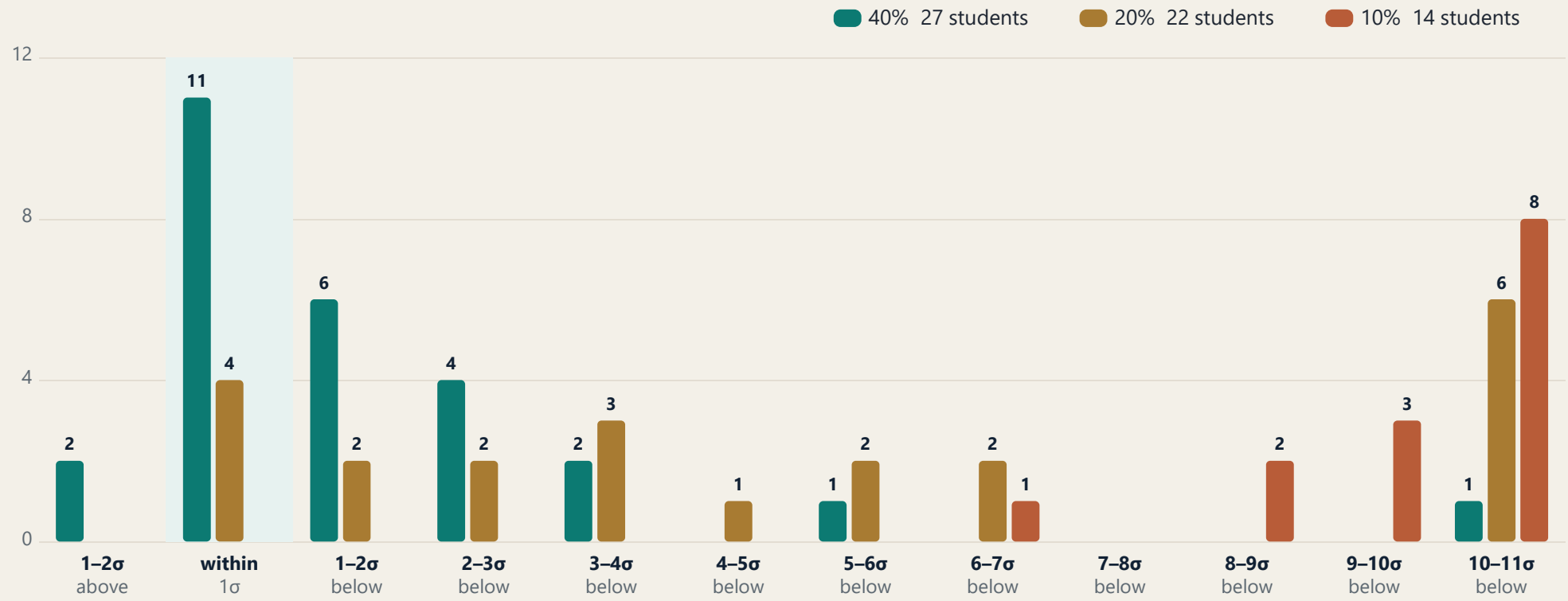
40% compression

RANK	ALIAS	SCORE
1	Gautam Mahale	50
2	Kislay Adtya Oj	48
3	Prakhar Paliwal	46
4	Baraneetharan	44
—	*Baseline*	*43.5 ± 4.12*
5	Aadeshveer Singh	42
5	Amit Tiwari	42

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Leaderboards Stats

Each bar is one standard deviation of the base score, 4.12 points. Base mean is 43.5.

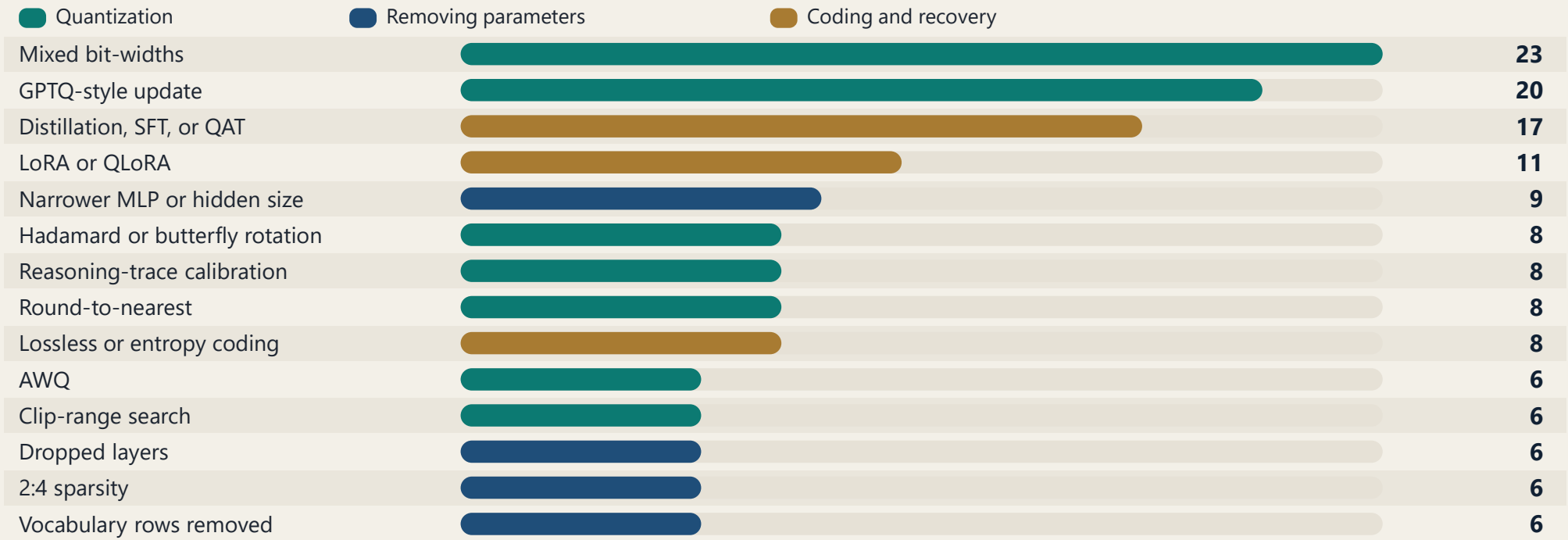


Best Track 1 score per student. Within 1σ is 39.4 to 47.6. Scores of 0 to 2.3, including the zeros, sit in the 10–11σ bar.

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Which techniques were tried?

Report counts, including zeros. A student can appear in several bars. Group quantization is omitted.



ALSO TRIED · FEWER THAN SIX REPORTS

- 5 Ternary levels
- 4 Rare-token bits
- 3 Shorter traces
- 2 OBS
- 1 EoRA
- 5 AutoRound
- 3 GPTAQ
- 2 QuIP
- 1 Minitron
- 1 NF3
- 4 Vector quant.
- 3 Fisher
- 2 QTIP
- 1 HQQ
- 1 NVFP4
- 4 Wanda
- 3 SVD
- 2 SparseGPT
- 1 OmniQuant

Who figured out it was HMMT?

ONLY ONE REPORT

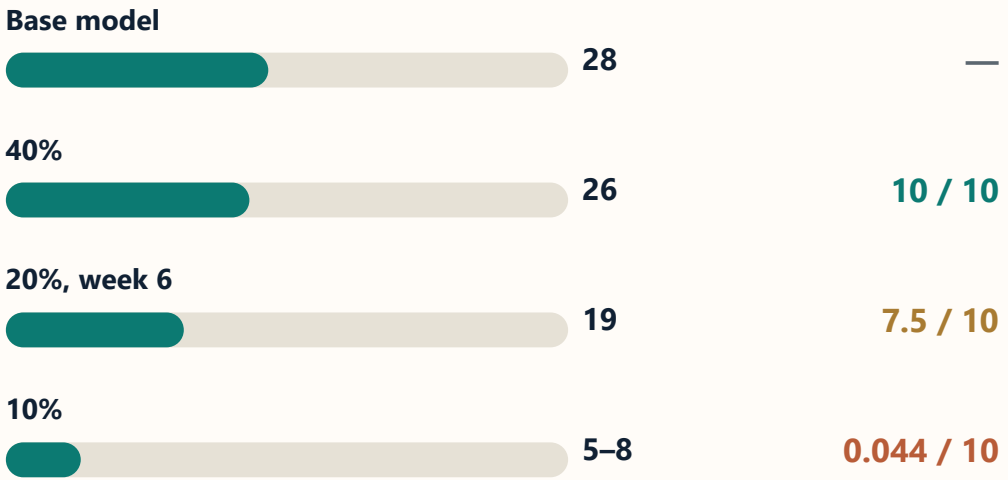
Sri Surya Varma Datla

On 2 October a 160-problem holdout was not tracking official scores.

Selection moved to 60 answer-only problems from HMMT February 2024 and February 2025.

"HMMT tracks the official scores well."

2 October	60	2024 + 2025
Switch date	Answer-only problems	HMMT February



USED HMMT LOCALLY, AND DID NOT CALL IT THE HIDDEN EVAL

Advay Bapat · Kislay Oj · Prakhar Paliwal · Rahul Chhabra · Siddharth Reddy

No other report mentions HMMT.

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Who fine-tuned?

A loss has to update weights. GPTQ and AWQ calibration is not included.

17 OF 27 REPORTS — **10 DID NOT**



TRAINED

Aadeshveer Singh
Amit Tiwari
Deeptanshu Malu
Deevyanshu Malu
Devansh Rathor
Gautam Mahale
Keshaw Ranjan
Kislay Oj
Malay Kedia
Prakhar Paliwal
Ratan Kokal
Rishi Vishwakarma
Rishit Aggarwal
Rudrapratap Chinhara
Siddharth Reddy
Sri Surya Varma Datla
Sumedh S S

DID NOT

Advay Bapat
Ansh Savla
Anurag Borkar
Apoorv Goyal
Baraneetharan A
Bhavesh Gyanchandani
Parthiv MEV
Praveen
Rahul Chhabra
Vedant Saini

Partial runs still count. Devansh: 10% and 20% only. Malay: scales only. Sri Surya: 10% only. Sumedh: 20% only.

Ratan's submitted checkpoint is still the base model. Anurag's rank-32 term is a stock GPTQ adapter. Bhavesh and Ansh say retraining is needed, and do not run it.

Named training sets

The dataset is what was named. The method is how it was used.

OpenR1-Math-220k KL distillation, all three sizes, 600 steps	**Deeptanshu Malu**
GSM8K, MATH-500, MMLU-Pro, AIME 2024 SFT + distillation. The 40% model is not fine-tuned	**Devansh Rathor**
MATH-500, GSM8K, AIME 2024 and 2025 LoRA after pruning, then QLoRA. 2,993 examples	**Rishi Vishwakarma**
GSM8K and Hendrycks MATH Scales only. Algebra, precalculus, intermediate algebra	**Malay Kedia**
NuminaMath-CoT Full fine-tune, 15 epochs, on 256 samples, then GPTQ	**Rudrapratap Chinhara**
OpenMath and OmniMath SFT. The compressed files are still the base model	**Ratan Kokal**

Which model was the teacher?

10 billion, 1.31 billion, and 90 million name the teacher. They are not training runs.

THE USUAL TEACHER

Qwen3.5-4B

Rishit distills both the 20% and 10% models from this checkpoint. On the 20% model the prompts are NVIDIA Nemotron prompts; the teacher that writes the answers is still the 4B model. Sri Surya's 10% distillation starts from the same model.

AN EXTRA SOURCE OF TRACES

Qwen3.5-9B

Sri Surya adds 9B traces in a second stage of the 10% run. Prakhar's later 10% stage mixes the teacher's solutions with answers from a 9B model. A 27B distillation is only proposed, and was not run.

EVERYONE ELSE

Where a report distills or matches a teacher and does not name another model, the teacher is the Qwen3.5-4B checkpoint being compressed. Sri Surya does not train the 40% or 20% models.